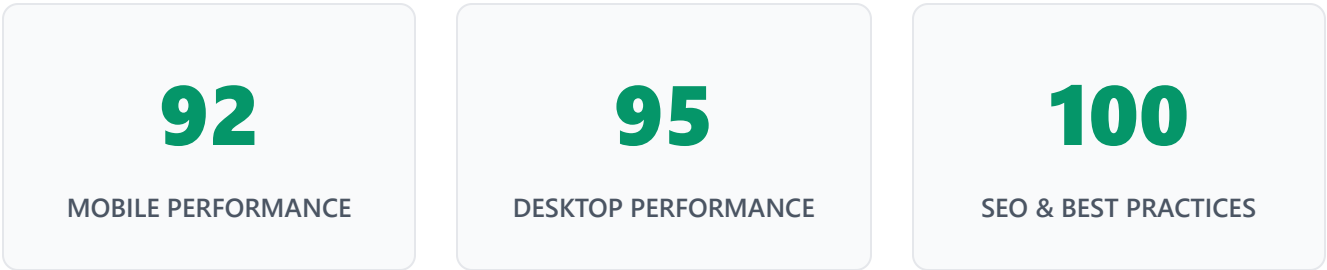


High-Performance Static Web Architecture: Case Study

Benchmark audit of a modern content portal built with **Astro v7**, **Tailwind CSS v4**, and lightweight modular motion engineering.

1. VERIFIED GOOGLE PAGESPEED INSIGHTS SCORES



2. CORE WEB VITALS BENCHMARK (LAB DATA)

Metric	Result	Standard Target	Status
First Contentful Paint (FCP)	0.8s	< 1.8s	GOOD
Largest Contentful Paint (LCP)	1.2s	< 2.5s	GOOD
Total Blocking Time (TBT)	0 ms	< 200 ms	PERFECT
Cumulative Layout Shift (CLS)	0.00	< 0.10	PERFECT

3. ARCHITECTURAL DECISIONS DRIVING SPEED

- ✓ Zero-JS Islands: Shipped 0 KB runtime framework JS on static informational views.
- ✓ Tailwind v4 CSS-First Theme: Zero PostCSS runtime overhead; inlined critical styles.
- ✓ Gzip/Brotli Payload Compression: Reduced raw HTML payload from 210 KB to 28 KB.
- ✓ Deferred DOM Reflows: Eliminated cold-load getBoundingClientRect layout thrashing.
- ✓ Edge CDN Deployment: Instant global TTFB < 80ms via Cloudflare edge routing.

